

COURSE TITLE : IMAGE PROCESSING
COURSE CODE : 3102
COURSE CATEGORY : B
PERIODS/ WEEK : 5
PERIODS/ SEMESTER : 75
CREDIT : 5

TIME SCHEDULE

MODULE	TOPIC	PERIODS
1	Prepress basics	20
2	Digital prepress	20
3	Screening	20
4	File formats and proofing	15
TOTAL		75

COURSE OUTCOME :

G .O.	ON THE COMPLETION OF THE STUDY OF THIS MODULE STUDENT WILL BE ABLE :
1	To understand the basic prepress operations.
2	To Comprehend the concept of color and color reproduction basics
3	To Comprehend the digital prepress work
4	To understand screening and dot shapes
5	Comprehend different file formats and their use
6	Understand proof and proofing methods

MODULE I PREPRESS BASICS

1.1.0 To understand the basic prepress operations

- 1.1.1. To explain prepress operations
- 1.1.2. To identify different types of images and graphics
- 1.1.3. To define art work components
- 1.1.4. To explain different prepress components

1.2.0 To comprehend the concept of color and color reproduction basics

- 1.2.1. To define the color
- 1.2.2. To describe color perception
- 1.2.3. To identify factors affecting color perception
- 1.2.4. To explain color theories
- 1.2.5.

MODULE II DIGITAL PREPRESS

2.1.0 To Comprehend the digital prepress work

- 2.1.1 To define digital prepress
- 2.1.2 To define input and output device operations

- 2.1.3 To explain digital data input through different mediums
- 2.1.4 To distinguish raster and vector images
- 2.1.5 To explain use of different types of software used in prepress

MODULE III SCREENING

3.1.0 To understand screening and dot shapes

- 3.1.1 To explain the need of screening
- 3.1.2 To classify types of screening
- 3.1.3 To explain IS and RT screening
- 3.1.4 To define HQS screening
- 3.1.5 To list types of IS and RT screening with their screen angle and relative screen frequency.
- 3.1.6 To explain resolution and quality parameters
- 3.1.7 To identify color shift and moiré
- 3.1.8 To explain trapping

MODULE IV FILE FORMATS AND PROOFING

4.1.0 Comprehend different file formats and their use

- 4.1.1 To identify different file formats
- 4.1.2 To explain the use of different file formats
- 4.1.3 To describe production of different file formats with different application software

4.2.0 Understand proof and proofing methods

- 4.2.1 To define proofs
- 4.2.2 To identify different types of proofs
- 4.2.3 To state progressive proof
- 4.2.4 To define OK sheet
- 4.2.5 To explain different proofing methods
- 4.2.6 To describe modern proofing techniques
- 4.2.7 To identify use of digital printers as proofing devices

CONTENT DETAILS

MODULE I

Prepress operations: typesetting, copy editing, graphic creation, proofing, proofreading, page layout preparation, manufacturing of Digital files/masters (PDF etc), imposition, screening & color separation (Digital), Manufacturing of masters.

Images and Graphics : Types of images – line images , continuous images, half tone images. Highlight area, middle tone area, shadow area. Color images, monochrome images.

Basic Elements of art work – text, image, and graphics.

Prepress components: – Computer, software, printers, scanners and Master preparation equipments.

Introduction to DTP.

Color: Wavelength of colors. Color perception, subjective color perception, factors effecting color perception. Color theories – additive, subtractive. Application of color theories in print production.

MODULE II

Definition of digital prepress. Digital prepress work flow. General Work flow with computer input and output devices. Data accruing through – input devices (text, pictures), application software (Photoshop, CorelDraw, etc), internet.

Vector and Bitmap images. Differentiate raster and vector.

Different types of prepress software : - Typesetting software . Image editing software - Pixel based image editing software, Vector graphics editing software, Page Layout Preparation software.

Imposition introduction. Factors affecting imposition. Imposition through software.

MODULE III

Definition of screening. Need of screening . Screening – Types of screening, IS and RT screening. Types of IS and RT screening with their screen angle and relative screen frequency. HQS screening.

Resolution and Quality: Pixel, PPI, DPI, LPI, screen angle, screen angle direction, Half tone dot shapes. Moiré effect. Color shift.

Different dot shapes and their use. Relation between dot shapes and LPI/resolution.

Trapping - definition, types – spread, choke.

MODULE IV

Different File formats- their features, and uses in printing – JPEG/JPG, TIFF, PNG, GIF, BMP, PDF, JDF, EPS, PSD, CRD, DOC/DOCX, PPT/PPTX, FLASH-SWF, ZIP, HTML, QXD, PMD, INDD, SLA/SCD, TXT, ODT, ICC/ICM, XML. Creating/saving in different file formats with different application programs (eg, PSD – Photoshop, CDR – Corel draw, DOCX – MS Word etc). Differentiate Vector file formats and raster file formats.

Definition of proof. Need of proof. Types of proof – soft proof, hard proof. Imposition (Layout) proof, color proof, screen proof. Advantages and disadvantages of hard and soft proof. Definitions of Prepress proof and press proof. Definition and use of progressive proof. Definitions of OK/ PASS sheet. Contract proof – definition.

Proofing methods: digital proofing – ink jet and laser printers. Digital halftone proof. Color and black and white materials. Current industry proofing methods.

Reference :		
Author	Title	Publishers
Dr R K Molla	Electronic colour separation	R.K. Printing & Publishing company, USA
Michael Bernard, John Peacock.	Handbook of printing and production	Springer Science & Business Media
Adams J.M, Faux,D.D and Rieber L.J	Printing Technology	Delmar Publishers, NewYork
Helmut Kipphan	Hand book of print media	Springer Science & Business Media
Reid Anderson	Exploring Digital Prepress : 1 edition	Delmar Cengage Learning